QUIZ REVIEWER

PART 1: Overview of Software and Hardware

🔹 **Hardware**

- Refers to the **physical parts of a computer**.

Overview of Software and Hardwa...

- Also called the **machinery or equipment**.
- The **motherboard** is the most important component because it controls and powers the computer.

Overview of Software and Hardwa...

✅ Examples: keyboard, monitor, CPU, mouse

🔹 **Software**

- Software is a set of **instructions/programs** that tell hardware what to do.

Overview of Software and Hardwa...

- Created by developers and must be compatible with the OS and CPU.
- Compatibility issues may occur between different operating systems.

✅ Examples: Windows, MS Word, mobile apps

🔹 **Top Emerging Technologies (Know the concepts!)**

- Generative AI
- Quantum Computing
- Edge AI
- Extended Reality (AR/VR/MR)
- Biotech & Gene Editing
- Autonomous Systems & Robotics
- Renewable Energy Tech
- 6G & Next-Gen Connectivity
- Human-Machine Interfaces

- Cybersecurity Enhancements

📌 Key idea: These technologies are shaping the future of IT.

🔹 **Generative AI**

- Creates **new content** (text, images, music).
- Used in art, programming, and business optimization.

🔹 **Quantum Computing**

- Uses **quantum bits (qubits)**.
- Much faster than traditional supercomputers.
- Useful in cryptography and drug discovery.

🔹 **Edge AI**

- Processes data **locally on devices** instead of the cloud.
- Benefits:
    - lower latency
    - better privacy
    - real-time decisions

🔹 **Trends in Software Development (7)**

- Artificial Intelligence
- Cross-Platform Tools
- Blockchain
- IoT
- Continuous Delivery & Deployment
- Progressive Web Apps
- Low-Code Development

📊 **PART 2: Requirement Analysis Techniques**

🔹 **Requirement Analysis (Definition)**

- Process of **identifying, eliciting, analyzing, specifying, validating, and managing stakeholder needs**.

Requirement Analysis Techniques

- Bridges business goals and technical solution.
- Helps reduce:
    - project risk
    - scope creep
    - budget overrun

🔹 **BPMN (Business Process Modeling Notation)**

- Graphical representation of business processes.
- Uses standard objects:

✅ Flow objects
✅ Connecting objects
✅ Swim lanes
✅ Artifacts

📌 Purpose: easy communication of processes.

🔹 **UML (Unified Modeling Language)**

- Standard modeling language for software systems.
- Used for:
    - specification
    - visualization
    - documentation

Two Types:

- Behavioral model → what the system does
- Structural model → what the system consists of

Flowchart

- Visual representation of **sequential flow of activities**.

- Easy for non-technical people to understand.
- Can show parallel processes.

🔹 **Data Flow Diagram (DFD)**

Shows how data moves in a system.

Components:

- Process
- Data Flow
- Data Store
- Terminator (external entity)

📌 Types:

- Logical DFD → activities
- Physical DFD → infrastructure

🔹 **Role Activity Diagram (RAD)**

- Similar to flowchart but focuses on **roles**.
- Shows responsibilities and interactions.
- Supports parallel activities.

🔹 **Gantt Chart**

- Graphical schedule of project tasks.
- Vertical axis → tasks
- Horizontal axis → time/duration
- Used for planning and tracking.

🔹 **Gap Analysis**

- Compares **current state vs desired state**.
- Steps:
    - Review system

    - Develop requirements
    - Compare
    - Identify implications
    - Recommend actions

🧠 **PART 3: Requirements Analysis and Modeling**

🔹 **Importance**

- One of the **most important skills** for a business analyst.
- Project success depends heavily on it.
- ...

🔹 **Main Tasks in RAM**

- Understand business requirements
- Decompose requirements
- Categorize requirements
- Model requirements

🔹 **Basic Objective**

- Understand customer needs.
- Translate them into technical requirements developers can use.

🔹 **What is a Requirement?**

- A **capability needed by the user** to solve a problem or achieve an objective.

📌 Goal: build software that meets real customer needs on time and within budget.

🔹 **Four Major Activities**

1. **Eliciting requirements**
    - Gather information from users
2. **Analyzing requirements**

    - Check for ambiguity or conflicts
3. **Requirements modeling**
    - Document using diagrams, user stories, etc.
4. **Review and retrospective**
    - Evaluate and improve process

🔹 **Business vs Software Requirements**

Business Requirements

- Focus on **WHY** the project exists
- Based on business goals
- High-level and clear

Software Requirements

- Focus on **WHAT** the system must do
- Detailed system functions
- Derived from business requirements